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VFMol: A Discrete Flow Matching Variational Autoencoder for Molecular Graph Generation

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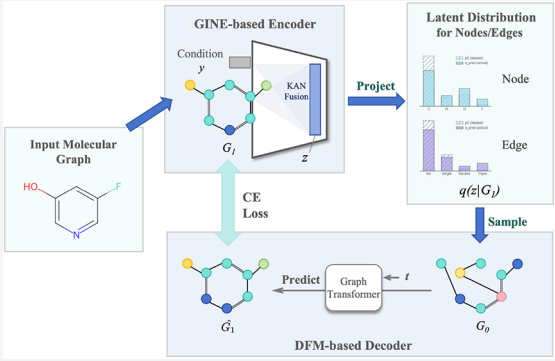
ABSTRACT: Molecular graph generation is a key task in drug discovery, aiming to efficiently identify novel compounds with desired properties. While variational autoencoders (VAEs) excel at latent space modeling and discrete flow matching (DFM) enables efficient continuous-time sampling, existing approaches still face critical limitations. VAE decoders struggle with permutation invariance and suffer from one-shot generation bottlenecks, whereas DFM models often rely on a fixed prior initialization that lacks adaptability to specific molecular structures. To address these issues, we propose VFMol, a novel framework that synergistically integrates personalized VAE latent space modeling with the efficient stepwise sampling mechanism of DFM in the discrete space. Specifically, the encoder learns a posterior distribution tailored to each input graph as the generation starting point, thereby enhancing both structural fidelity and diversity. Moreover, we introduce a lightweight property-guided framework based on KAN and classifier-free guidance, enabling conditional generation without auxiliary property predictors. Experiments on two widely used molecular data sets demonstrate that VFMol achieves state-of-the-art performance in terms of molecular structural quality and property controllability, verifying its generality and effectiveness.



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■ INTRODUCTION

In drug discovery and chemical biology, de novo molecular design seeks to generate novel compounds with desired pharmacological activities and favorable physicochemical properties from first principles.[1] The vastness of the accessible chemical space renders exhaustive exploration through traditional experimental screening or combinatorial enumeration.[2] Consequently, the development of systematic and efficient computational strategies for navigating this vast chemical space has emerged as a critical challenge. Recently, driven by the availability of large-scale molecular data sets and rapid advancements in deep learning, data-driven models have emerged as a crucial approach to accelerate drug discovery.[3,4]

Molecular graph generation models aim to learn the underlying probability distribution of molecular structures from training data, enabling efficient exploration of the vast chemical space.[5] From the generative modeling perspective, these approaches can be broadly categorized into four classes, including variational autoencoders (VAEs),[6] generative adversarial networks (GANs),[7] normalizing flows,[8] and diffusion models.[9] Among them, diffusion-based methods tailored to discrete graph structures[10−12] have achieved remarkable performance by formulating generation as a stepwise denoising process in the discrete state space. For instance, DIGRESS[10] introduces a Markovian noising process that perturbs graphs through stochastic node and edge modifications, and trains a graph transformer to reverse this corruption, effectively transforming graph generation into a classification task. However, such models typically rely on iterative sampling governed by stochastic differential equations (SDEs), which incur high computational cost and results in slow inference.[13] The flow matching (FM) framework[14] provides a new perspective for continuous generative modeling. Instead of learning stochastic dynamics, FM trains a model to predict a continuous vector field that deterministically maps a prior distribution to the data distribution via ordinary differential equations (ODEs). This approach enables stable training, dramatically accelerates sampling, and explicit control over generation trajectories.[15] Recent work has extended FM to discrete domains,[16,17] demonstrating its efficacy in molecular graph generation. CatFlow[18] formulates FM within a variational inference framework, GGFlow[19] integrates optimal transport with edge-enhanced graph Transformers to build a discrete FM model, and Defog[20] decouples the FM training and sampling phases. Despite these advantages, discrete FM models often employ a fixed prior

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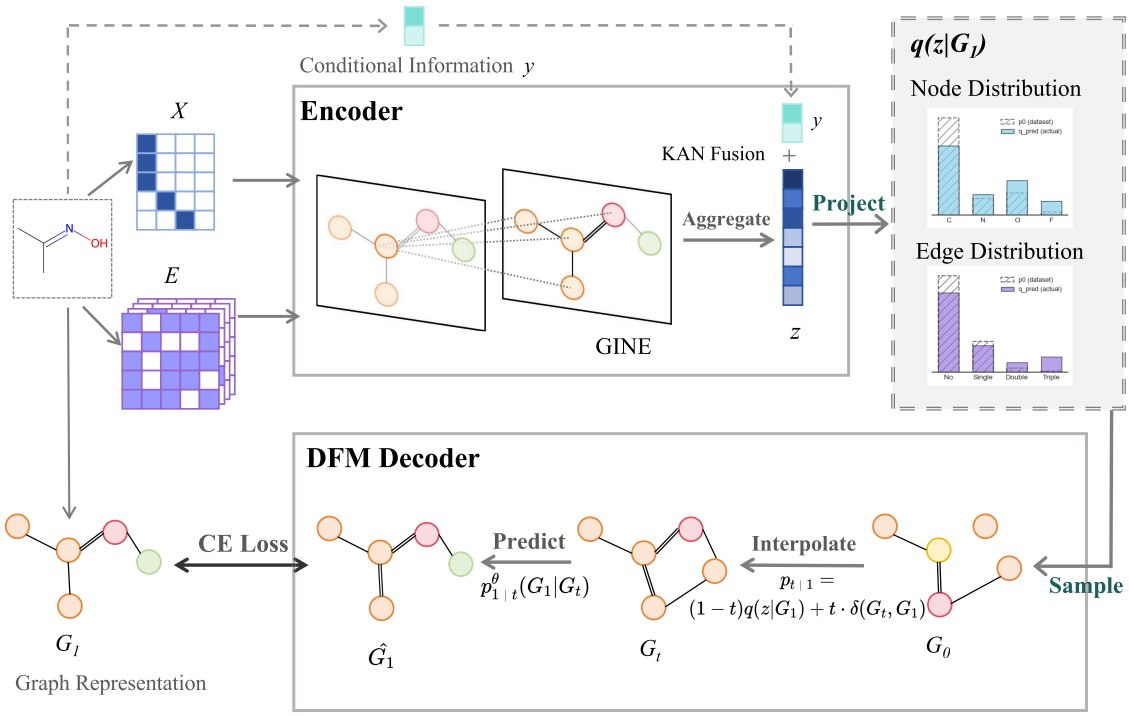

Figure 1. Overview of VFMol. VFMol consists of a GINE-based encoder and a DFM-based decoder. During training, the encoder extracts node and edge features from the input molecular graph $G_1 = (X, E)$ and aggregates them to obtain a latent variable z. This variable is projected into a latent distribution $q_\phi(z|G_1)$, which serves as a customized initialization for the decoder. The decoder linearly interpolates q_ϕ and p_1 to generate a noisy graph G_t, from which a neural network $p^\theta_{1|t}(G_1|G_t)$ predicts the marginal distribution of the denoised graph $\hat{G}_1$. Node and edge predictions are processed independently to accommodate their distinct structural properties. For conditional generation tasks, the model incorporates additional conditional information y, indicated by blue blocks and dashed arrows.

distribution. This lack of input-specific initialization limits generation diversity and hinders the capture of fine-grained local topological features.

In contrast, classical VAEs[21,22] adopt a joint encoder-decoder architecture, where the encoder maps a molecular graph to a molecule-specific latent representation, and the decoder reconstructs the graph from this latent code. GraphVAE,[21] employs a decoder that outputs a probabilistic fully connected graph of a predefined maximum size and then aligns it to the ground-truth graph via graph matching, which is suitable for small-scale graphs. Set2GraphVAE[22] first generates a node set, and then learns edge connectivity with a Set2Graph module to refine latent representations. While VAE-based approaches feature simple structures and efficient training, they encounter challenges in graph decoding. First, permutation invariance is difficult to preserve, leading to sensitivity to node ordering, often necessitating extra graph-matching operations.[21] Moreover, as decoders adopt one-shot whole-graph reconstruction from latent variables, it is difficult to balance efficiency and fidelity, especially for large and complex graphs.

These observations reveal a natural complementarity. The fixed-prior limitation of discrete FM can be mitigated by the input-adaptive latent space of VAEs, while the decoding bottleneck of VAEs can be overcome by the stepwise generation process of FM. Hybrid models combining VAEs with other frameworks have begun to emerge in molecular graph generation. Graphusion[23] combines Variational Graph Autoencoders with latent space diffusion models. It maps high-dimensional discrete graph data to a lower-dimensional continuous latent space and performs diffusion generation within this latent space. GF-VAE[24] employs a lightweight flow model as the VAE decoder, mapping from the encoder's posterior Gaussian distribution, thereby reducing parameter overhead. The hybrid methods rely on continuous latent space modeling, which does not fully align with the discrete structural characteristics of molecular graphs. The inherent smoothness of continuous latent spaces conflicts with the discrete "node−edge" topology of molecules, leading to potential information loss and blurred discrete attributes.

To address these limitations, we propose VFMol, a hybrid molecular graphs generation framework that deeply integrates VAE and discrete Flow Matching in the discrete state space. VFMol combines the latent space learning capability of VAEs with the efficient stepwise sampling of discrete FM. Specifically, the framework is built on two core modules: a GINE-based encoder that learns a posterior distribution tailored to each input molecular graph, serving as a customized starting state, and a DFM-based decoder that incrementally samples the target graph structure from this initialization. By initializing the generation process with a structure-aware latent code, VFMol eases the learning task for the decoder, enhances generation quality, and improves sample diversity. Furthermore, to enable property-conditioned generation, we design a lightweight conditional framework that leverages Kolmogorov−Arnold Networks (KAN)[25] to capture complex correlations between molecular properties and latent variables, combined with classifier-free guidance for precise property control without external property predictors. This design ensures both efficiency and flexibility, addressing targeted drug discovery demands.

Our contribution can be summarized as

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- We propose VFMol, a novel molecular graph generation framework that integrates VAEs with discrete Flow Matching in the discrete space. This hybrid approach preserves the personalized latent space modeling capability of VAEs while leveraging the stepwise sampling of Flow Matching, thereby balancing generation quality and computational efficiency.
- We design a lightweight property-guided framework by introducing KAN to capture complex correlations between molecular properties and latent variables. Combined with classifier-free guidance, our method enables efficient property-conditioned generation without requiring external property predictors.
- Experimental results on two benchmark data sets demonstrate that VFMol achieves state-of-the-art performance in both unconditional and conditional molecular generation, validating its effectiveness and generalizability.

■ MATERIALS AND METHODS

The Overview of VFMol

We present VFMol, a hybrid molecular graph generation framework that integrates variational inference with discrete Flow Matching to enable high-fidelity and diverse molecule synthesis. As illustrated in Figure 1, VFMol formulates molecular graph generation as a structured reconstruction task and consists of two core components with an extended conditional module: (1) a GINE-based encoder for learning molecule-specific latent representations, (2) a discrete Flow Matching (DFM)-based decoder for stepwise graph refinement, and (3) a lightweight conditional generation module for property-guided synthesis. This framework synergistically combines the input-adaptive representation power of VAEs with the efficient, stepwise refinement capability of discrete Flow Matching, enabling VFMol to generate chemically valid, diverse and property-targeted molecules with improved fidelity and computational efficiency.

Specifically, a molecule can be modeled as an undirected graph G, where nodes correspond to atoms and edges to chemical bonds. A graph with n nodes is formally defined by its node feature matrix $X \in \mathbb{R}^{n\times d_x}$ and its edge feature tensor $E \in \mathbb{R}^{n\times n\times d_e}$. The features for both nodes and edges are represented as one-hot vectors indicating atom and bond types, respectively. Our objective is to learn the underlying data distribution $p(G)$ and generate novel molecular graphs that preserve the structural and chemical properties of real molecules. For the input molecular graph G, the encoder maps G to a posterior distribution $q_\phi(z|G)$ over a discrete latent space. This latent space is defined as a compact, structure-aware space that represents global molecular topology and chemical composition. The initial distribution in the generative process refers to the starting point from which molecular graphs are iteratively refined by the decoder. Rather than relying on fixed global priors, we use the learned posterior to sample a customized initial state that captures molecule-specific structural information. During training, the DFM-based decoder takes this initial state and learns to reconstruct the original graph by predicting the underlying vector field that governs the denoising process in the discrete graph space. Specifically, the decoder employs a graph Transformer to iteratively refine node and edge attributes through a deterministic ODE-based sampling process derived from the Flow Matching objective. The entire model is trained end-to-end using a composite loss function that combines a reconstruction term (based on discrete Flow Matching) and a regularization term (from the variational encoder), ensuring both generation quality and latent space coherence.

VAE-Encoder

The objective of the encoder ϕ is to map the graph G to a posterior distribution $q_\phi(z|G)$ over a discrete latent space. This posterior serves as a customized initial state that encodes global structural information unique to the input molecule. By replacing the fixed prior commonly used in standard generative models, this approach enhances the model's ability to capture instance-specific structural patterns.

Specifically, we adopt a three-layer GINEConv[26] as the encoder backbone. GINEConv effectively integrates node features with adjacent edge attributes, enabling richer message passing and demonstrating strong representational power for complex molecular graphs. For each layer k ($k = 0, 1, 2$), the embedding h_u^k of node u is updated as

$$h_u^{k+1} = f_\Theta\left[(1+\epsilon)\cdot h_u^k + \sum_{v\in N(u)} \mathrm{ReLU}(h_v^k + e_{u,v}^k)\right] \tag{1}$$

where $f_\Theta[\cdot]$ denotes a multilayer perceptron (MLP), and $N(u)$ represents the set of neighboring nodes of node u. $e_{u,v}^k$ represents the edge between node u and node v at layer k, and ϵ is a learnable scalar parameter.

The edge features $e_{u,v}$ are dynamically updated based on the embeddings of their two end points, enabling richer context-aware representations

$$e_{u,v}^{k+1} = \mathrm{MLP}([h_u^k, h_v^k, e_{u,v}^k]) \tag{2}$$

At the final layer $T = 3$, all node representations h_u^T are aggregated into a global graph representation h_G. Specifically, we adopt a combination of global average pooling and global max pooling to capture both holistic and salient structural information

$$\begin{aligned} h_G &= \mathrm{Readout}(\{h_u^T\}_{u\in V}) \\ &= \mathrm{Concat}(\mathrm{MeanPool}(\{h_u^T\}), \mathrm{MaxPool}(\{h_u^T\})) \end{aligned} \tag{3}$$

The resulting graph-level vector h_G is passed through a MLP to produce the latent variable $z = \mathrm{MLP}(h_G)$. Subsequently, z is projected onto spaces matching the cardinalities of node and edge types, yielding the categorical distributions for node and edge types

$$q_x = \mathrm{Proj}_x(z) \in \Delta^{dx},\ \ q_e = \mathrm{Proj}_e(z) \in \Delta^{de} \tag{4}$$

where Δ^k denotes the k-dimensional probability simplex: $\Delta^k = \left\{p \in \mathbb{R}^k \middle| \sum_i^k p_i = 1,\ p_i \geq 0,\ \forall\, i\right\}$.

The resulting pair of distributions, constitutes our variational posterior $q_\phi(z|G_1) = (q_x, q_e)$, which replaces the standard Gaussian prior $N(0,I)$ used in conventional VAEs operating in continuous latent spaces. By aligning the prior with the empirical marginal distribution of the data set, our approach preserves the sparsity inherent in molecular graphs. Compared to fixed priors, this design enables more flexible modeling of structural regularities and provides structure-aware guidance during decoding.

To address potential instability from encoder underfitting during early training, we introduce an interpolation strategy balancing the prior p_{ref} and the learned posterior $q_\phi(z|G_1)$. The interpolated distribution p_0 is formulated as

$$p_0 = (1-\lambda)p_{\mathrm{ref}} + \lambda q_\phi(z|G_1) \tag{5}$$

where the interpolation coefficient $\lambda \in [0, 1]$ increases linearly with the training epochs. Empirical results demonstrate that this strategy improves training stability and accelerates convergence.

The DFM model samples the initial state G_0 from the interpolated distribution p_0, which serves as the starting point for the reverse generative process.

Jensen−Shannon Regularization. To guide the encoder in learning a latent distribution $q_\phi(z|G_1)$ that close to the reference prior distribution $p(z) = p_{\mathrm{ref}}$, we introduce a Jensen-Shannon(JS) divergence-based regularization term to constrain the latent space

$$\mathcal{L}_{\mathrm{JS}} = \mathrm{JS}(q_\phi(z|G_1), p_{\mathrm{ref}}) = \mathrm{JS}(q_x, p_x) + \mathrm{JS}(q_e, p_e) \tag{6}$$

where the JS divergence is defined as

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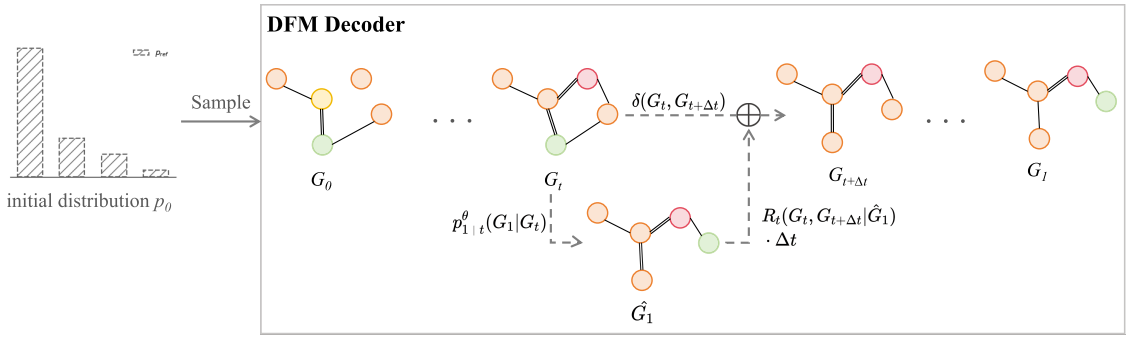

Figure 2. Sampling process of VFMol. Starting from the initial distribution, at each time step t, the neural network $p^{\theta}_{1|t}(G_1|G_t)$ predicts a denoised graph $\hat{G}_1$, computes its rate matrix R_t, and simulates the transition to the next state $t + \Delta t$. This process is iterated until the graph converges to the target molecular structure.

$$\mathrm{JS}(q, p) = 0.5 \times \mathrm{KL}(q, m) + 0.5 \times \mathrm{KL}(p, m),$$
$$m = 0.5 \times (q + p) \tag{7}$$

Compared to KL divergence, JS divergence is more suitable for our task. In early training stages, the learned posterior distribution $q_\phi(z|G_1)$ often differs significantly from the reference prior p_{ref}. In such cases, KL divergence imposes harsh penalties on regions with low support under the target distribution, potentially leading to divergence and gradient instability. In contrast, JS divergence is symmetric and introduces an intermediate mixture distribution m, which softens the penalty on nonoverlapping regions and provides a smoother optimization objective. This helps progressively align the latent representation with the reference prior.

DFM-Decoder

The DFM-based decoder reconstructs the original graph from an initial state by learning the vector field governing the denoising trajectory in discrete graph space. Specifically, we adopt Discrete Flow Matching (DFM)[16] as our VAE decoder to model the transformation from initial distribution p_0 to the target data distribution $p_1 = p_{\mathrm{data}}$ over molecular graphs. Since molecular graphs comprise discrete node and edge types, we define the flow over a discrete state space, constructing a stochastic process over $t \in [0, 1]$ that gradually transforms a randomly sampled graph $G_0 \sim p_0$ into a molecular graph G_1 following the target distribution p_1.

Structure-Aware Initialization. In standard DFM, the initial distribution p_0 is typically a fixed prior, such as a uniform distribution or the empirical marginal distribution of the data set. However, such a global and static initialization ignores the structural diversity among molecules, limiting the flexibility and adaptability of the generative process. To address this limitation, VFMol introduces a structure-aware initialization strategy. Instead of a fixed p_0, we use the latent distribution $q_\phi(z|G_1) = (q_x, q_e)$ learned from the encoder. Specifically, node and edge types are independently sampled respectively to construct the initial graph G_0.

This strategy offers two key advantages. First, unlike fixed priors, our encoder provides instance-specific initialization by mapping each input G_1 to a unique latent representation z, creating a customized starting point that better aligns with target structure and reduces decoder burden. Second, while traditional approaches inject diversity solely through decoder randomness, our VAE framework learns a rich latent distribution $q_\phi(z|G_1)$ for each input, enabling the initial graph itself to encode meaningful diversity. This distribution-to-distribution paradigm better captures molecular graph complexity.

Probability Path and Rate Matrix. In the DFM framework,[16] the generative process is described jointly by the probability path p_t and the rate matrix R_t. The probability path describes the distribution of the graph state G_t at each time step $t \in [0, 1]$, evolving from the initial distribution p_0 toward the target distribution p_1. The infinitesimal evolution of state probabilities is governed by the rate matrix $R_t \in \mathbb{R}^{d_Z \times d_Z}$, where d_z denotes the number of discrete graph states. Each entry $R_t(i, j)$ represents the instantaneous transition rate from state i to state j at time t.

The transition probability of the Markov jump process over an infinitesimal time interval d_t satisfies

$$p_{t+dt|t}(G_{t+dt}|G_t) = \delta(G_t, G_{t+dt}) + R_t(G_t, G_{t+dt})\cdot dt \tag{8}$$

where $\delta(i, j) = \mathbb{I}\{i = j\}$ denotes the Kronecker delta. Here, G_t denotes the graph structure state at time t, including all discrete node and edge states.

To ensure conservation of total probability mass $\sum_j p_t(G_j|G_t) = 1$, the diagonal elements of the rate matrix R_t must satisfy

$$R_t(i, i) = -\sum_{j \neq i} R_t(i, j), \quad \forall\, i \leq d_Z \tag{9}$$

Since the probability path p_t lacks a closed-form solution, DFM employs the conditional path $p_{t|1}(G_t|G_1)$ constructed through linear interpolation

$$p_{t|1}(G_t|G_1) = \mathrm{Cat}((1 - t)\cdot p_0 + t\cdot\delta(G_t, G_1)) \tag{10}$$

where $\mathrm{Cat}(\cdot)$ denotes a categorical distribution over states. With G_1 fixed, this conditional distribution admits closed-form evaluation.

Following Campbell et al.,[16] we employ the rate matrix

$$R^*(G_t, G_{t+dt}|G_1) = \frac{\mathrm{ReLU}(\partial_t p_{t|1}(G_{t+dt}|G_1) - \partial_t p_{t|1}(G_t|G_1))}{Z_t \cdot p_{t|1}(G_t|G_1)}$$
$$\text{where } G_{t+dt} \neq G_t \tag{11}$$

where $\mathrm{ReLU}(x)$ ensures non-negativity, and the normalization factor $Z_t = |\{G_t: p_{t|1}(G_t|G_1) > 0\}|$. As shown in Appendix A, this simplifies to

$$R^*(G_t, G_{t+dt}|G_1) = \frac{\delta(G_{t+dt}, G_1)}{Z_t \cdot (1 - t)p_0}, \quad \text{where } G_{t+dt} \neq G_t \tag{12}$$

Training Objective. Since the posterior $p_{1|t}(G_1|G_t)$ is intractable, we train a neural network $p^{\theta}_{1|t}(G_1|G_t)$ to predict the marginal distribution of G_1 given noisy G_t. We adopt the graph transformer architecture from DiGress,[10] which employs multihead attention to simultaneously encode node, edge, and global features, capturing complex structural dependencies.

The training objective minimizes the negative log-likelihood, equivalent to a cross-entropy loss that decomposes into node and edge components

$$\mathcal{L}_{\mathrm{CE}} = -\mathbb{E}_{G_1 \sim p_1, t \sim U(0,1), G_t \sim p_{t|1}(G_t|G_1)}(\log p^{\theta}_{1|t}(G_1|G_t))$$
$$= \mathrm{CE}(x_1, p^{\theta}_{1|t}(x_1|x_t)) + \mathrm{CE}(e_1, p^{\theta}_{1|t}(e_1|e_t)) \tag{13}$$

Sampling. Consistent with DFM, VFMol leverages a Continuous-Time Markov Chain (CTMC)[27] to model the temporal evolution of graph states, a process governed by a rate matrix R_t. Specifically, generation initiates from $G_0 \sim p_0$ and transitions through intermediate

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states $\{G_t\}$, ultimately producing G_1 approximating data set samples. The overall sampling workflow is illustrated in Figure 2.

We approximate this process using Euler discretization. At each infinitesimal time step Δt, the next state $G_{t+\Delta t}$ is sampled from

$$G_{t+\Delta t} \sim p_{t+\Delta t|t}(G_{t+\Delta t}|G_t) \tag{14}$$

According to eq 8, this sampling distribution depends on the unconditional rate matrix

$$R_t(G_t, G_{t+\Delta t}) = \mathbb{E}_{p_{1|t}(G_1|G_t)} R(G_t, G_{t+\Delta t}|G_1) \tag{15}$$

We adopt an unbiased Monte Carlo approximation to replace the explicit computation of expectations over all conditional rate matrices. At any given time t, we first draw a sample $\hat{G}_1$ from the probability distribution $p_{1|t}(G_1|G_t)$ predicted by the graph transformer. Then, $\hat{G}_1$ is plugged into the rate matrix $R_t(G_t, G_{t+\Delta t}|\hat{G}_1)$ as an unbiased estimator of the true expectation. Finally, the next graph state $G_{t+\Delta t}$ is sampled from the rate-driven conditional path $p_{t+\Delta t|t}$

$$G_{t+\Delta t} \sim p_{t+\Delta t|t}(G_{t+\Delta t}|G_t) \approx \delta(G_t, G_{t+\Delta t}) + \Delta t \cdot R_t(G_t, G_{t+\Delta t}|\hat{G}_1) \tag{16}$$

Finally, to ensure chemical validity, we apply postcorrection mechanisms checking valence constraints and performing minimal interventions to correct structural violations.

In summary, the DFM-based decoder provides a powerful alternative to standard VAE decoders. By simulating CTMC dynamics, it generates graphs incrementally through local updates, avoiding the bottleneck of single-step generation from latent variables. This incremental approach enhances stability and reduces dependence on latent variable expressiveness. Furthermore, as detailed in the subsequent section, the DFM framework is inherently permutation-invariant, ensuring robustness to graph isomorphisms without requiring computationally expensive graph matching operations typical in traditional VAE models.

Loss Function and Training Framework

The VFMol model is trained end-to-end by optimizing a composite loss function that combines the reconstruction objective of the DFM decoder with the regularization objective of the VAE encoder

$$L = \mathrm{CE}(\hat{G}_1, G_1) + \beta \cdot \mathrm{JS}(q_\phi(z|G_1), p_{\mathrm{ref}}) \tag{17}$$

Following the β-VAE framework,[28] we introduce hyperparameter β to weight the regularization term, controlling the trade-off between graph reconstruction fidelity and latent space structure. In our experiments, we set $\beta < 1$ to avoid overly strict latent space constraints and to ensure the model can effectively express complex graph structures.The training and sampling processes are provided by Algorithm 1 and Algorithm 2.

Algorithm 1 Training Procedure of the model VFMol

Input: a graph $G_1 = (X_1, E_1)$, $p_{\mathrm{ref}} = (p_x, p_e)$
$H = \mathrm{GINE}(X_1, E_1)$ # Encoder
$z = \mathrm{MaxPool}(H)$
$q_\phi(z \mid G_1) = \mathrm{Project}(z)$ # Project to latent logits
Sample $t \sim U(0,1)$ # Decoder
Sample $G_t \sim p_{t|1}(G_t \mid G_1) = (1-t)q_\phi(z \mid G_1) + t \cdot \delta(G_t, G_1)$
$\hat{G}_1 = p^\theta_{1|t}(G_1 \mid G_t) = \mathrm{Graph\ Transformer}(G_t, t)$
$L = \mathrm{CE}(\hat{G}_1, G_1) + \beta \cdot \mathrm{JS}(q_\phi(z \mid G_1), p_{\mathrm{ref}})$
optimizer.step(L)

Permutation Equivariance Analysis

Permutation invariance of node ordering is a fundamental property of graph-structured data. For a graph with n nodes, applying a node permutation $\pi \in S_n$ (where S_n denotes the symmetric group) can transform its adjacency matrix into multiple forms that represent the same semantic structure. To ensure stability and generalizability in graph generation, the model architecture must exhibit permutation equivariance while maintaining a permutation-invariant loss function.

Formally, permutation equivariance is defined as

$$\pi.f(x) = f(\pi.x), \quad \forall\, \pi \in S_n \tag{18}$$

This property ensures that the model produces structurally consistent outputs, so the output structure remains unchanged regardless of node ordering.

Proposition 1. The VFMol model architecture is permutation equivariant, and its loss function is permutation-invariant.

A direct consequence of this design is that the generative distribution learned by VFMol is exchangeable, meaning the probability of generating a specific graph is independent of its node labeling.

Proposition 2. The generative distribution $p(G)$ of the VFMol model is exchangeable. For any permutation π, the node feature matrix X and adjacency matrix A satisfy

$$P(X, A) = P(\pi^T X, \pi^T A \pi) \tag{19}$$

The formal proofs for Proposition 1 and Proposition 2 are provided in Appendix B.

Conditional Generation: Molecular Property Targeting

To enable the generation of molecules with specific, desirable chemical properties, we extend VFMol with a conditional generation framework based on classifier-free (CF) guidance.[29] This approach allows the model to directly generate molecules that adhere to a target property value y (e.g., $\log P$ or QED score) without requiring an external property prediction model.

The training process follows the same overall workflow as unconditional generation, with the key difference lying in the encoding stage, where Kolmogorov−Arnold Networks (KAN)[25] are introduced to fuse latent variables z with conditional property information y. Specifically, KAN replaces the conventional MLP to capture more complex nonlinear dependencies between y and z, thereby accurately modeling the interaction between molecular properties and latent features. This enables the computation of a variational posterior distribution $q_\phi(z|G_1,y)$, which more effectively incorporates conditional information into latent representation learning and yields a customized initialization state that better satisfies property constraints.

Here, KAN follows a lightweight two-layer design, where each layer consists of a set of learnable univariate spline transformations with additive aggregation. In contrast to MLPs that model interactions implicitly through stacked linear layers, KAN explicitly parametrizes the nonlinear relationships between latent variables and conditional properties through adaptive basis functions. In conditional molecular generation, the dependency between molecular properties and latent representations is typically highly nonlinear and exhibits cross-dimensional heterogeneity. By decomposing multivariate interactions into combinations of learnable univariate nonlinear mappings, KAN enables more accurate modeling of how molecular properties influence the latent space, thereby improving conditional controllability during the generation process.

Algorithm 2 Sampling Procedure of the model VFMol

Require: Decoder θ, $p_{\mathrm{ref}} = (p_x, p_e)$, time interval Δt, node number distribution p_n
Init: $t = 0$
Sample node number $n \sim p_n$
Sample initial graph $G_0 \sim p_{\mathrm{ref}}$
while $t < 1$ **do**
  Sample $\hat{G}_1 \sim p^\theta_{1|t}(G_1 \mid G_t) = \mathrm{Graph\ Transformer}(G_t, t)$
  Sample $G_{t+\Delta t} \sim p_{t+\Delta t|t}(G_{t+\Delta t} \mid G_t) = \delta(G_t, G_{t+\Delta t}) + R_t(G_t, G_{t+\Delta t}) \cdot \Delta t$
  Update $t \leftarrow t + \Delta t$, $G_t \leftarrow G_{t+\Delta t}$
end while
return $G_1 = (X_1, E_1)$

Since the conditional denoising network $p^\theta_{1|t}(G_1|G_t,y)$ may tend to ignore the condition y, we adopt the conditional dropout technique,[29] which randomly replaces y with a placeholder y^* with probability ρ. This forces the model to learn both the property-conditioned distribution $p_\theta(G_1|G_t,y)$ and the unconditional distribution $p_\theta(G_1|G_t,y^*) = p_\theta(G_1|G_t)$.

During sampling, the property-conditioned transition probability can be expressed as

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Table 1. Molecular Generation Results on Two Datasets QM9 and ZINC250k[a]

dataset	method	Val. ↑	Val. w/o corr↑	unique↑	novelty ↑	FCD↓
QM9	GraphVAE		61.0	40.9	80.5	2.476
	PS-VAE	100		67.3	52.3	1.259
	Moflow	100	96.17	**98.65**	94.72	4.467
	DiGress	100	98.19	96.67	25.58	0.095
	GruM	100	**99.69**	96.90	24.15	0.108
	HGGT	100	99.22	95.65	24.01	0.40
	TarMGDif	100	98.19	55.93	97.59	0.350
	Caflow	100	93.19	79.20	74.56	5.018
	Defog	100	99.3	96.3	38.32	0.12
	Ours	100	99.54 ± 0.05	96.83 ± 0.06	37.63 ± 0.08	**0.093 ± 0.004**
ZINC250k	PS-VAE	100		99.7	99.7	6.320
	Moflow	100	63.11	**99.99**	**100**	20.93
	DiGress	100	92.13	99.96	**100**	3.606
	GruM	100	**98.65**	99.97	99.98	2.257
	HGGT	100	92.87	99.97	99.83	1.93
	TarMGDif	100	80.52	99.95	99.89	5.862
	Caflow	100	43.21	99.28	**100**	26.11
	Defog	100	95.24	**99.99**	99.94	2.145
	Ours	100	96.48 ± 0.23	**99.99 ± 0.01**	99.98 ± 0.01	**1.871 ± 0.038**

[a]The first four metrics are reported in percentage format.

$$p_{t+\Delta t|t}(G_{t+\Delta t}|G_t, y) = \delta(G_t, G_{t+\Delta t}) + R_t(G_t, G_{t+\Delta t}|y)\cdot\Delta t \tag{20}$$

where the conditional rate matrix is computed as

$$\begin{aligned} R_t(G_t, G_{t+\Delta t}|y) &= E_{p_{1|t}(G_1|G_t,y)} R(G_t, G_{t+\Delta t}|G_1, y) \\ &= \sum_{\widehat{G}_1} R(G_t, G_{t+\Delta t}|\widehat{G}_1)\cdot p^{\theta}_{1|t}(\widehat{G}_1|G_t, y) \end{aligned} \tag{21}$$

To steer generation toward desired property values, we construct a guidance term combining conditional and unconditional distributions. The enhanced sampling direction is defined as

$$p^{\theta}_{1|t}(\hat{G}_1|G_t) + s\cdot(p^{\theta}_{1|t}(\hat{G}_1|G_t, y) - p^{\theta}_{1|t}(\hat{G}_1|G_t)) \tag{22}$$

where $s \geq 0$ denotes the guidance scale controlling conditional influence strength. We empirically set $s = 2$ in our experiments.

■ RESULTS

Experimental Setup

Datasets. We evaluate VFMol's generation capabilities on two standard benchmarks: QM9[30] and Zinc250k.[31] QM9 comprises small molecules with up to 9 atoms from 4 types. Zinc250k is a larger data set containing molecules with up to 38 atoms from 9 types. For both data sets, edges represent four bond types (no bond, single, double, triple). Aromatic bonds in the raw data are systematically converted to alternating single and double bonds using the RDKit Kekulize function[32] during preprocessing.

Baselines. We benchmark VFMol against nine representative models spanning several generative paradigms:

- VAE-based: GraphVAE,[21] PS-VAE.[33]
- Flow-based: MoFlow.[34]
- Diffusion-based: DiGress,[10] GruM,[35] TarMGDif.[36]
- Flow Matching-based: CatFlow,[18] Defog.[20]
- K^2-tree representation based: HGGT.[37]

We reproduce results for CatFlow and Defog on Zinc250k. Results for MoFlow and DiGress are sourced from the reproductions in Holderrieth and Erives.[13] All other baseline results are cited from their original publications. Due to its scalability limitations, GraphVAE is evaluated only on QM9.

Evaluation Metrics. The performance of these methods are quantified using five widely used evaluation metrics:

- **Validity (Val.):** The percentage of generated molecules that are chemically valid according to RDKit.
- **Validity w/o correction (Val. w/o corr.):** The validity rate before any posthoc correction. This metric directly reflects the model's ability to learn chemical rules.
- **Uniqueness:** The percentage of unique molecules among all valid generated samples.
- **Novelty:** The percentage of valid, unique molecules that do not appear in the training set.
- **FCD (Fréchet ChemNet Distance):** Measures the distributional similarity between generated molecules and the test set; a lower score indicates higher fidelity.

Implementation Details. We train VFMol for 300 epochs with a batch size of 512 for QM9 and 256 for Zinc250k. The generation process starts by sampling the initial noisy graph from the marginal empirical distribution p_{ref}, followed by 1000 steps of iterative sampling through the decoder to obtain the final molecular graph. Following standard practice, we randomly split data into training (80%), testing (10%), and validation (10%) sets. For batch processing, all molecular graphs are padded to the maximum number of atoms in the data set. We generate 10,000 molecular samples for each data set and evaluate them using the above metrics. All experiments are repeated three times, and the results are reported as averages and standard deviations to reflect model stability, with the best results highlighted in bold and the second-best underlined.

Molecular Graph Generation

To comprehensively evaluate the molecular generation capability of VFMol, we compare it against the above nine representative generative models on the two benchmark data sets QM9 and Zinc250k. As shown in Table 1, VFMol demonstrates superior performance across critical metrics on both data sets. Notably, VFMol achieves the lowest FCD scores (0.093 on QM9, 1.871 on Zinc250k), outperforming the second-best model HGGT by 3%. This metric directly

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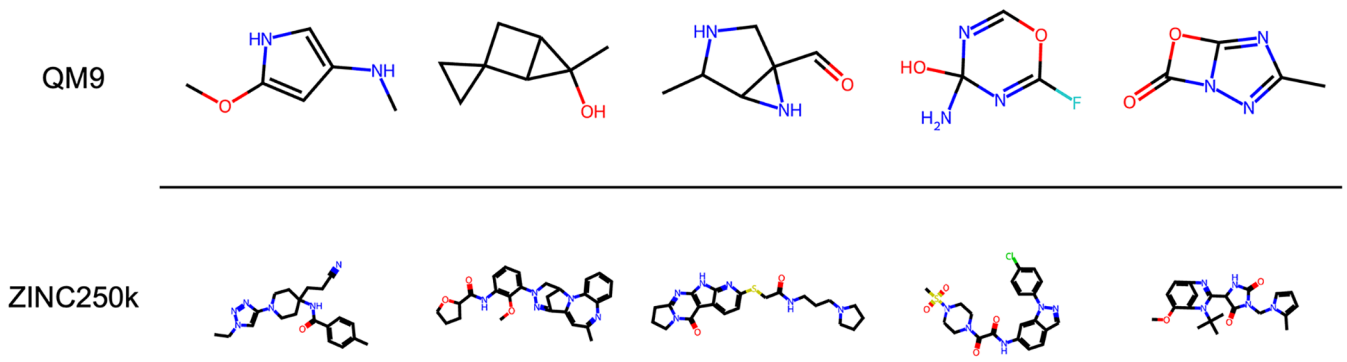

Figure 3. Visualization of the generated samples from the data sets QM9 and ZINC250k.

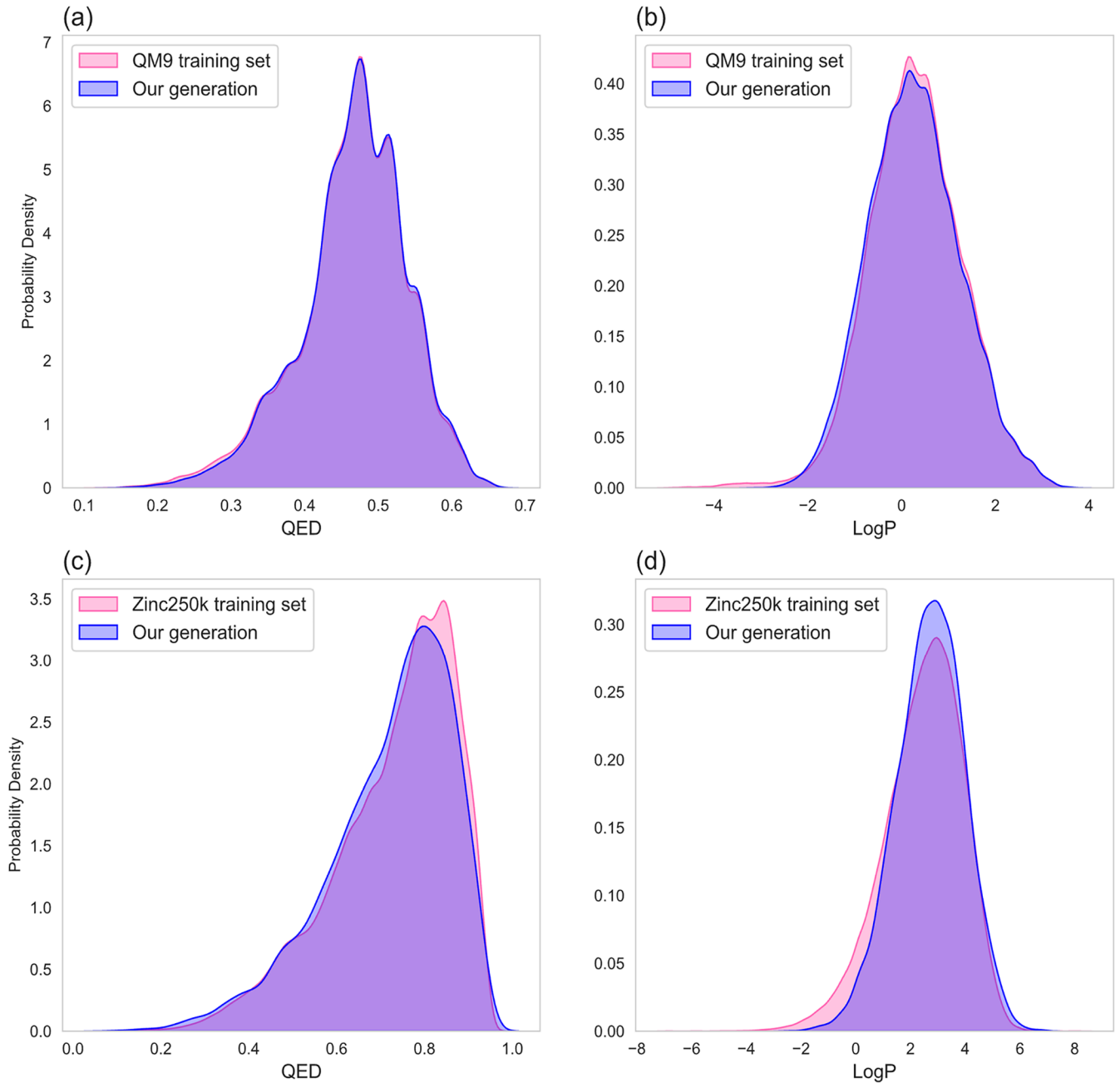

Figure 4. Comparison of QED and Log *P* distributions between training sets and the generated molecules of VFMol. (a) QED distribution on QM9. (b) Log *P* distribution on QM9. (c) QED distribution on Zinc250k. (d) Log *P* distribution on Zinc250k.

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quantifies distributional similarity, confirming the ability of VFMol to accurately capture real molecular distributions. Furthermore, VFMol excels in generating chemically valid structures without postprocessing, achieving raw validity rates of 99.54% on QM9 and 96.48% on Zinc250k. This highlights its capacity to learn the inherent rules of chemical structure validity.

It is worth noting that we do not emphasize the novelty metric for QM9. As discussed in previous studies proposed by Vignac and Frossard,[22] QM9 is a small molecule data set constrained by chemical rules, generating molecules outside the training set may not be reasonable and could indicate a deviation from chemical validity. On the Zinc250k data set, the uniqueness and novelty metrics are close to 100% for all models with minimal variance. Therefore, validity and FCD are more indicative of model performance and serve as the most critical indicators of performance.

The performance, particularly the low FCD scores, is further corroborated by a qualitative analysis of the generated molecules' chemical properties. As illustrated in Figure 4, the distributions for the QED and log P from VFMol-generated samples closely match those of the training set. This alignment confirms that our model accurately captures the underlying property distributions of the data. Figure 3 provides visual examples of the high-quality molecules produced, further validating our model's effectiveness and generalizability. Collectively, these findings demonstrate that VFMol achieves an excellent balance between distributional fidelity, structural integrity, and sample diversity.

Conditional Molecular Generation

Further, we evaluate conditional generation capabilities of VFMol on the Zinc250k, aiming to generate molecules with specific target properties. As previous studies proposed by Ninniri et al.,[38] we select log P and QED as the conditional properties. We randomly select 100 molecules from the test set and used their log P and QED values as distinct conditional targets y. For each condition, we generated 1000 sample molecules. Performance is primarily evaluated using the Mean Absolute Error (MAE) between the generated molecules' properties and the target value, with lower MAE indicating superior conditional control. All property calculations are performed using the RDKit library[32] to obtain the target property values.

The model trains for 500 epochs using classifier-free guidance, where conditional values are replaced with placeholder $y^* = -1$ at probability $\rho = 0.1$ to enable both conditional and unconditional generation. The guidance coefficient $s = 2$ strengthens property control. For the KAN-based fusion module, we adopt a two-layer KAN configuration. Each KAN layer employs cubic B-spline basis functions with a grid size of 10. We benchmark against DiGress,[10] a state-of-the-art molecular generation model, to assess conditional control performance.

Table 2 demonstrates that VFMol consistently outperforms DiGress in conditional generation. For log P, VFMol reduces the MAE by nearly 50% relative to DiGress (0.42 vs 0.83). The advantage is even more pronounced for QED, where VFMol attains MAE of 0.05 compared to DiGress's 0.14, a 64% reduction in error. Moreover, the validity of the generated molecules, without postcorrection, exceeds 92%, which, though slightly lower than the Unconditional model, demonstrates that the model can effectively maintain the chemical validity of the molecules while learning additional attribute constraints. This highlights the model's ability to balance property control and structural validity. These results demonstrate that VFMol can more precisely and effectively control the chemical properties of the generated molecules, indicating stronger property control capabilities in conditional generation tasks. Figure 5 presents the molecules generated via conditional sampling, further validating the generation performance of the model.

Table 2. Conditional Generation Performance on the Dataset Zinc250k for Log P and QED Property

	Log P		QED	
method	MAE ↓	Val. w/o corr↑	MAE ↓	Val. w/o corr↑
unconditional	1.52 ± 0.12	96.48	0.15 ± 0.01	96.48
DiGress	0.83 ± 0.08	77.6	0.14 ± 0.01	84.5
ours	**0.42 ± 0.02**	**93.86**	**0.05 ± 0.01**	**92.19**

Parameter Sensitivity Analysis

To further investigate the role of the guidance scale s in conditional molecular generation, we conduct a sensitivity analysis under the log P property constraint. The guidance scale controls the strength of classifier-free guidance, balancing the influence between conditional and unconditional generation during sampling. In this experiment, we evaluate five commonly used values, $s \in \{1, 1.5, 2, 2.5, 3\}$.

As shown in Figure 6, different values of s lead to distinct trade-offs between property accuracy and generation quality. When $s = 2$, the model achieves the lowest MAE, indicating the most accurate control over the target log P property while maintaining high molecular validity. Larger values of s slightly increase MAE and decrease validity, implying that overly strong guidance may overemphasize the conditional signal and disrupt the generative dynamics. In contrast, when $s = 1$, the unconditional component is completely removed, causing the model to degenerate into a purely conditional generator. This results in a higher MAE, highlighting the importance of retaining an unconditional component to stabilize generation and improve robustness.

Overall, the sensitivity analysis demonstrates that the guidance scale s plays a critical role in conditional molecular generation. Empirically, $s = 2$ provides the best balance and is therefore adopted as the default setting in our experiments.

Ablation Study

To verify the contribution of each key component, we perform an ablation study. Specifically, we compare three different configurations, with results summarized in Table 3

- **Configuration 1:** Adopts the GINE encoder + Transformer decoder. While retaining the distribution learned by the encoder, this configuration is limited by the expressive capacity of the Transformer decoder, resulting in low uncorrected Validity and a significantly higher FCD.
- **Configuration 2:** Removes the encoder and directly trains the DFM decoder starting from the empirical data distribution $p_0 = p_{\mathrm{ref}}$, making the overall model similar to DeFog.[20] Notably, all metrics show significant improvement, demonstrating that the DFM decoder substantially enhances generation quality.
- **Full model:** Builds on the DFM decoder by replacing the original fixed reference distribution with the latent

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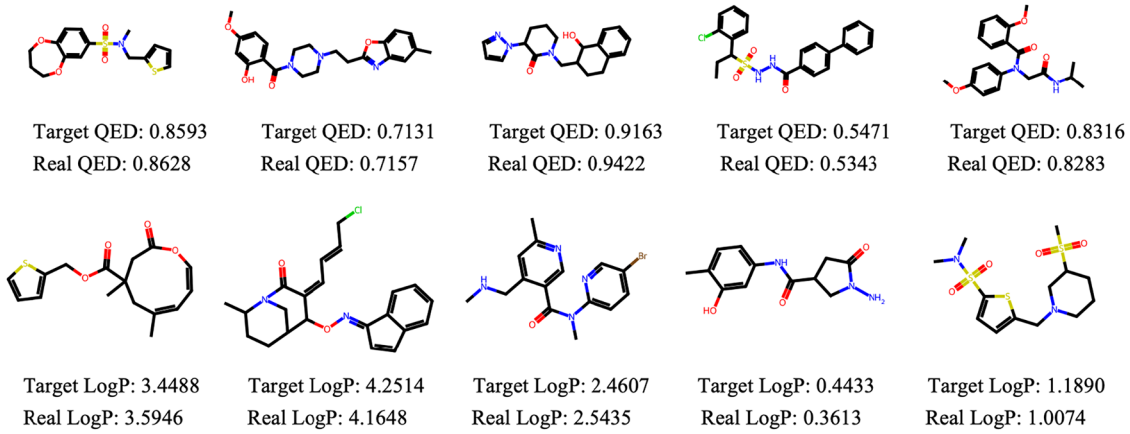

Figure 5. Representative molecules generated from conditional sampling on the data set Zinc250k, using models trained to control QED (top row) and Log P (bottom row). Target and real property values are shown below each structure.

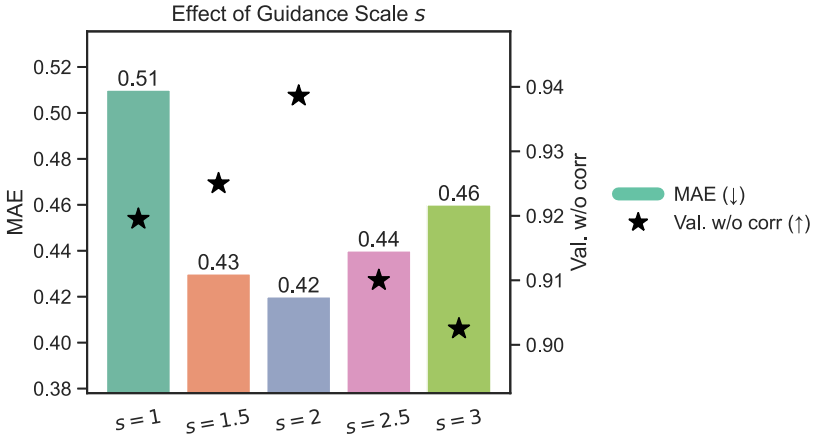

Figure 6. Effect of the guidance scale s on conditional molecular generation under log P property constraint. Bars indicate MAE with respect to the target log P value, while star markers denote Validity without correction.

Table 3. Performance of Different Model Variants on Molecular Generation[a]

		QM9		ZINC250k	
$p_0 = (q_x, q_e)$	DFM decoder	Val. w/o corr↑	FCD↓	Val. w/o corr↑	FCD↓
√	×	58.69 ± 0.08	2.06 ± 0.05	6.75 ± 2.09	21.16 ± 1.86
×	√	99.26 ± 0.09	0.120 ± 0.008	95.24 ± 0.08	2.145 ± 0.07
√	√	99.54 ± 0.05	0.093 ± 0.004	96.48 ± 0.23	1.871 ± 0.04

[a]Reported metrics include uncorrected validity (Val. w/o corr) and Fréchet ChemNet Distance (FCD) on the QM9 and ZINC250k datasets.

variable distribution learned by the encoder. With the combination of these two improvements, Validity and FCD reach optimal values.

These results reveal three key insights. First, the DFM decoder serves as the primary performance driver, substantially reducing distributional bias compared to traditional architectures. Second, the learned latent distribution provides approximately 20% FCD improvement over fixed priors on both data sets, demonstrating its effectiveness in capturing instance-specific patterns. Finally, the synergy between these components achieves optimal balance between validity and distributional fidelity, validating our architectural design.

■ CONCLUSION

This paper presents VFMol, a novel hybrid framework for molecular graph generation that integrates variational autoencoders with discrete flow matching (DFM). By synergistically combining these complementary components, VFMol addresses two key limitations of existing approaches, including the expressiveness bottleneck in traditional VAE decoders and the challenge of modeling permutation equivariance in graph-structured data. Specifically, the VAE encoder learns personalized latent distributions for each molecular graph, providing structure-aware initialization that enhances generation diversity. Meanwhile, the DFM decoder replaces conventional MLP architectures, enabling incremental graph refinement through continuous-time Markov chain dynamics. This hybrid architecture preserves the advantages of latent variable modeling while significantly improving both generation quality and computational efficiency. Experimental evaluation on benchmark data sets demonstrates that VFMol achieves state-of-the-art performance across multiple metrics. Notably, our method attains the lowest FCD values on both QM9 and ZINC250k data sets, indicating superior distribu-

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tional fidelity. Furthermore, the model maintains high validity rates while supporting flexible conditional generation for targeted molecular properties.

Our future work is to extend VFMol to broader graph generation tasks beyond molecular structures, evaluating its generalizability across diverse graph types and domains. Additionally, we aim to enhance the conditional generation framework by developing more sophisticated architectures for $q_\phi(z|G_1, y)$, potentially enabling multiproperty optimization and more precise control over generated molecular characteristics.

■ ASSOCIATED CONTENT

Data Availability Statement

The code and data sets can be accessed at the following GitHub repository https://github.com/DHUDBlab/VFMol. The repository includes the implementation of our proposed model and instructions for reproducing the experiments. Any additional data sets used in this study are either publicly available from their respective sources or provided in the repository with appropriate references.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.5c03005.

> Proof of the simplification of the rate matrix R_t^* (Appendix A); proof of the permutation equivariant property of the model (Appendix B) (PDF)

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Author Contributions

Y.G. conceptualized and structured the study, supervised the work, and edited the manuscript. J.S. developed and executed the method, conducted the experiments, analyzed the results, and wrote the manuscript. K.W. and G.X. conceptualized the study, and revised the manuscript. G.Z. managed the project, and revised the manuscript. All authors read and approved the final manuscript.

Notes

The authors declare no competing financial interest.

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